24-Hour Grand Finale Build Scope & Feasibility

A FOCUSED MVP TODAY. A STRONGER TOMORROW.

Build Today
Enable **Brighter Tomorrows**

1 24-HOUR BUILD SCOPE (MVP LOCK)

MVP LOCK
Focused. Realistic. Buildable.

- One exercise: grip-and-release / finger flexion-extension
- One flex sensor (index finger) + one EMG channel
- Automatic repetition counting with correct/incomplete classification
- Real-time haptic or visual feedback
- Basic session dashboard — live trace, rep count, session summary
- Nothing beyond this attempted until the MVP is solid

2 COMPONENT CHECKLIST

HARDWARE

- [x] ESP32 DevKit board
- [x] Flex sensor, 2.2" (x2)
- [x] EMG sensor module (MyoWare-class) + electrode pads
- [x] Base fabric glove for mounting
- [x] Vibration motor + LED indicators
- [x] Li-ion/LiPo battery + charging module
- [x] Breadboard/perfboard, resistors, wires, connectors

Total MVP cost:
~₹5,700

EMBEDDED SYSTEM

- [x] ADC sampling of flex (voltage-divider) + EMG (analog envelope) channels
- [x] Filtering, rectification, envelope/RMS extraction (firmware-side)
- [x] Peak/valley rep segmentation + per-finger calibration routine
- [x] BLE streaming to dashboard, with USB-serial as wired demo fallback
- [x] Feedback actuator driver (vibration motor / LED) on detected deviation

3 SCALABILITY (post-hackathon, no new clinical research needed)

- Personalized per-user calibration baselines
- Additional exercises + additional fingers/EMG channels
- IMU integration for compensatory-movement detection
- On-device inference — fully standalone device, no companion app
- Dedicated mobile app replacing the dashboard prototype

From a 24-hour MVP to real-world impact.